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Naïve Bayes Classifiers

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Ph.D. in Computer Science

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Why Learning NBC?

Course Evaluation Dataset

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail
Yes	Yes	Yes	???

Three features: Confident, Studied, and Sick

Two classes: Fail and Pass

class=? when Confident=Yes and
Studied=Yes
Sick=Yes

IRIS Flower Dataset

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

Two features: Length and Width (continuous)

Two classes: ‘0’ and ‘1’

class=? when Length=4.1 and Width=2.9

Objectives

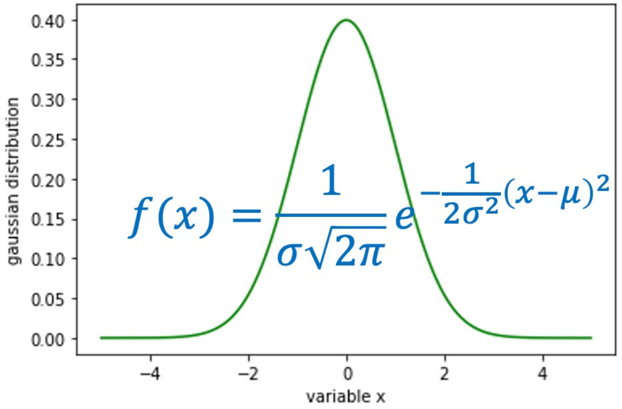
NBC for Discrete RV

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

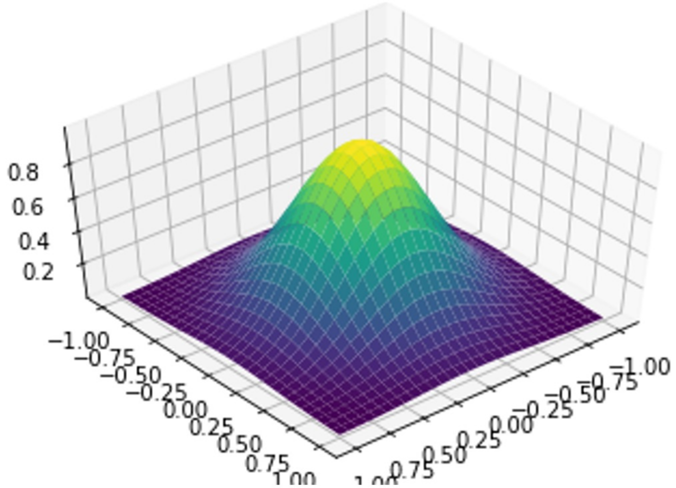
$$p(res = pass) = \frac{3}{6} = 0.5$$

$$p(res = fail) = \frac{3}{6} = 0.5$$

Gaussian Function

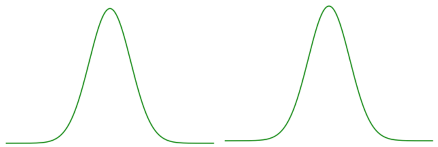


$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



NBC for Cont. RV

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1



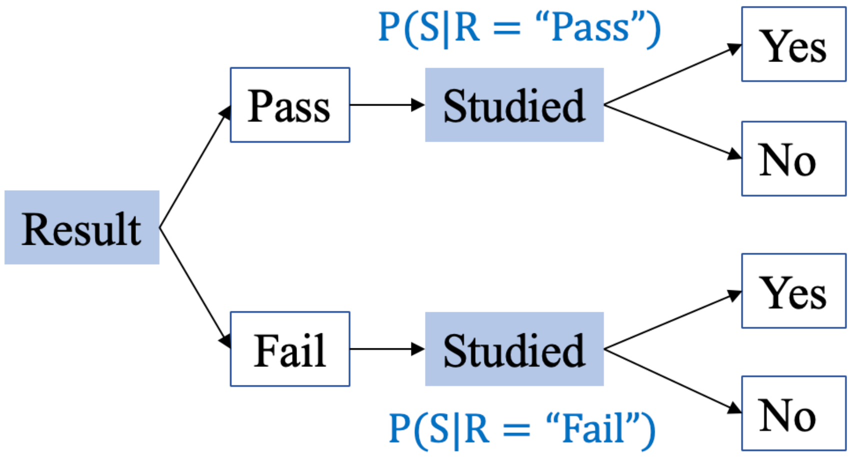
Outline

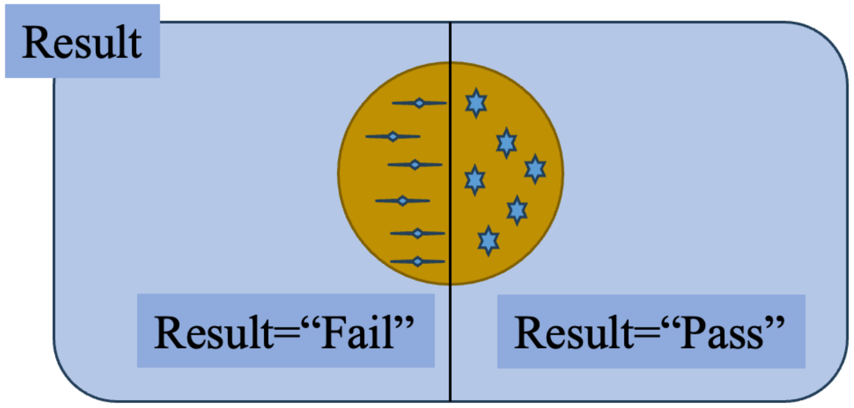
SECTION 1
Review

SECTION 2
NBC for Discrete RV

SECTION 3
Gaussian Function

SECTION 4
NBC for Continuous RV





Independent events

- Events A and B are independent if:

$$P(AB) = P(A)\ P(B)$$

- If A and B are not independent, they are dependent.

Example

A single fair dice is rolled. Let A={4} and B={2, 4, 6}.

Are A and B independent?

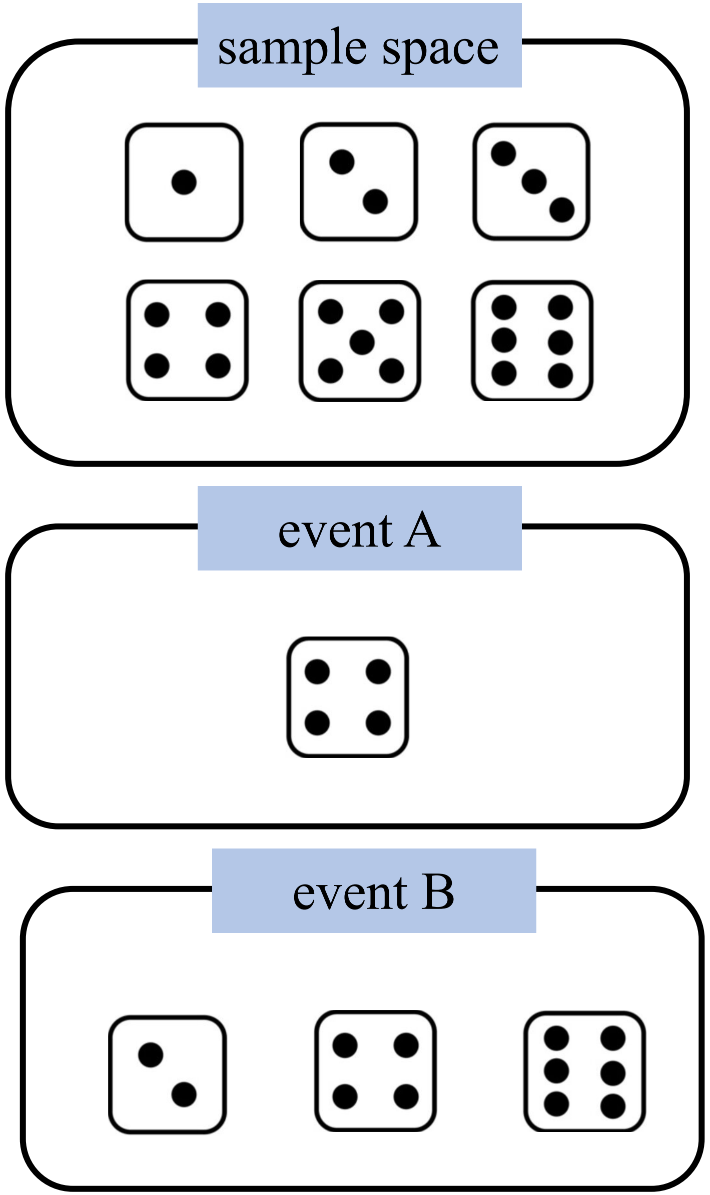
Compute: $P(A) = 1/6$

$P(B) = 1/2$

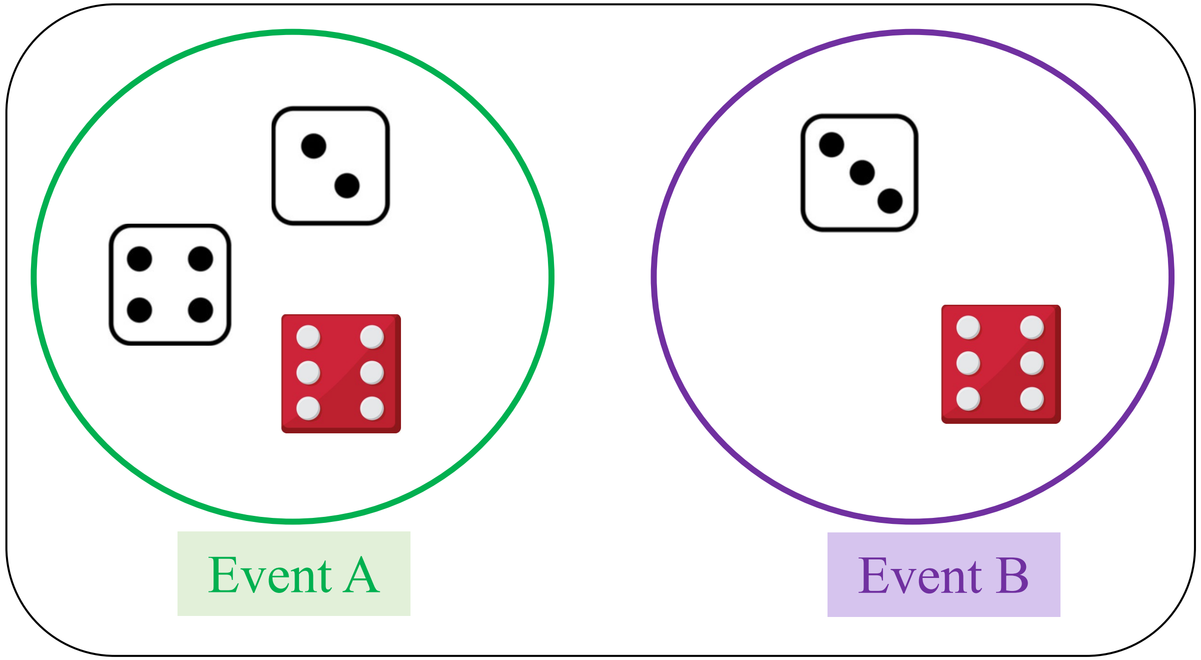
$P(A \text{ and } B) = 1/6$

Since $P(A)P(B) = (1/6)*(1/2) = 1/12 \neq P(A \text{ and } B) = 1/6$

=> Events A and B: not independent



Quiz 1: Cho A và B như hình dưới, hỏi có phải A và B là dependent hay independent?



Quiz 2: Phân tích mối quan hệ của các biến Studied và Result của data sau?

S	R
Studied	**Result**
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

R_1

R_2

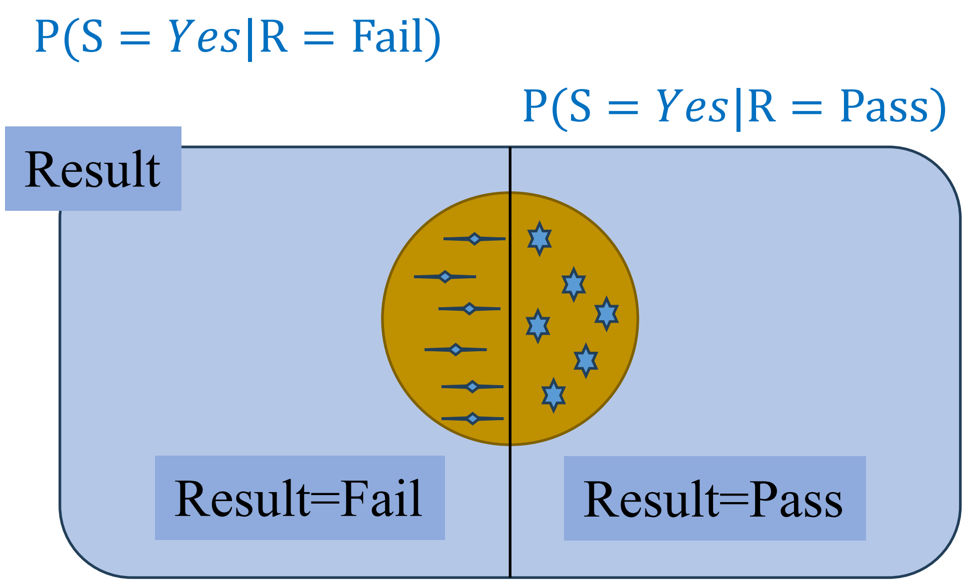
Total Probability Theorem

Let R_1 and R_2 – complete system of events.

Consider any event S such that S occurs only when one of the events R_1 and R_2 occurred

$$P(S = \text{Yes}) = P(S = \text{Yes} \cap R_1) + P(S = \text{Yes} \cap R_2)$$

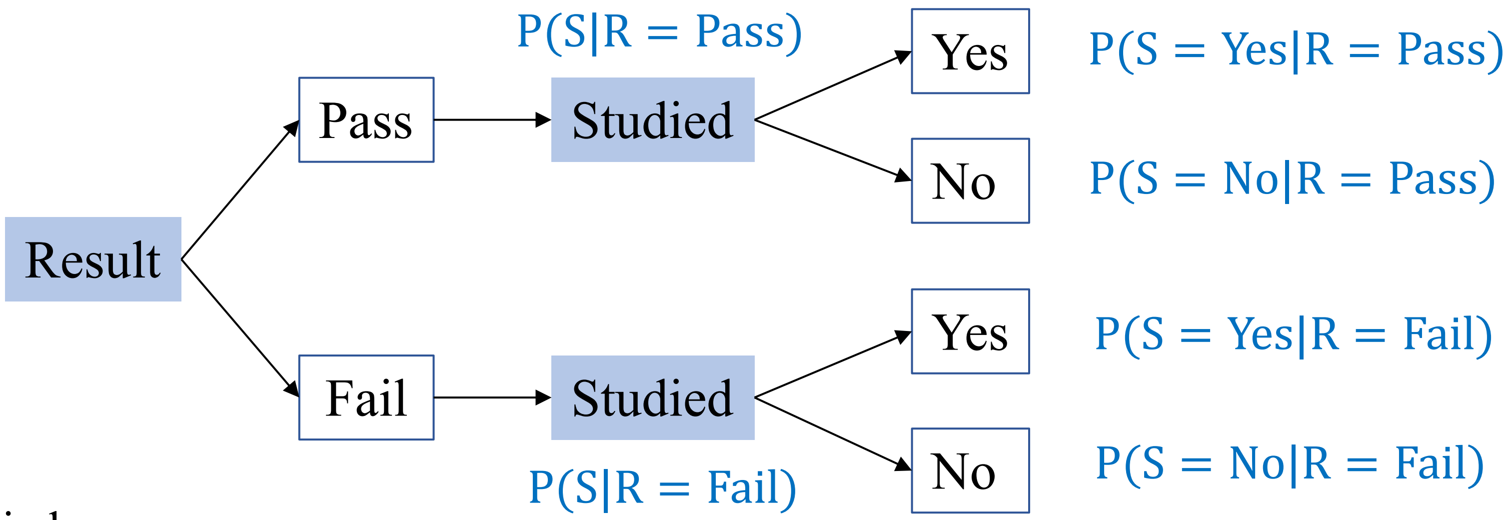
$$= P(R_1).P(S = \text{Yes}|R_1) + P(R_2).P(S = \text{Yes}|R_2)$$



Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

R_1

R_2



Assume Result is not dependent of Studied

Total Probability Theorem

Let C_1 and C_2 – complete system of events.

Consider any event W such that W occurs only when one of the events C_1 and C_2 occurred

$$P(W = 2.9) = P(W = 2.9 \cap C_1) + P(W = 2.9 \cap C_2)$$
$$= P(C_1).P(W = 2.9|C_1) + P(C_2).P(W = 2.9|C_2)$$

Width	Category
3	0
3.2	0
3.1	0
3.6	0
3.9	0
3.4	0
3.2	1
3.1	1
2.3	1
2.8	1
2.8	1
3.3	1

C_1

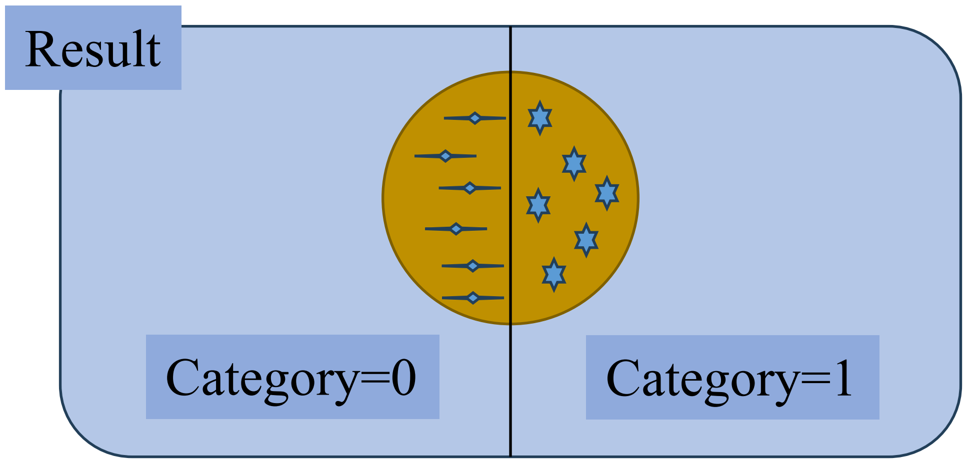
C_2

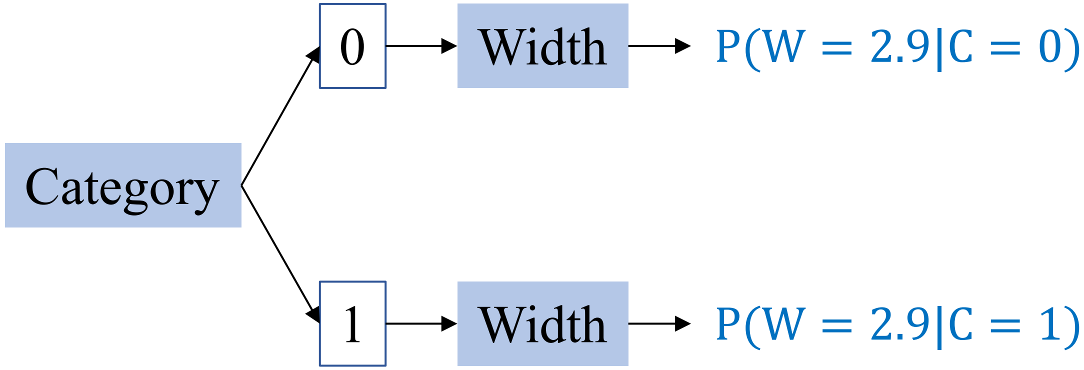
Two classes: ‘0’ and ‘1’

class=? when Width=2.9

Assume Category is not dependent of Width

$P(W = w|C = 0)$ $P(W = w|C = 1)$





Bayes' Rule

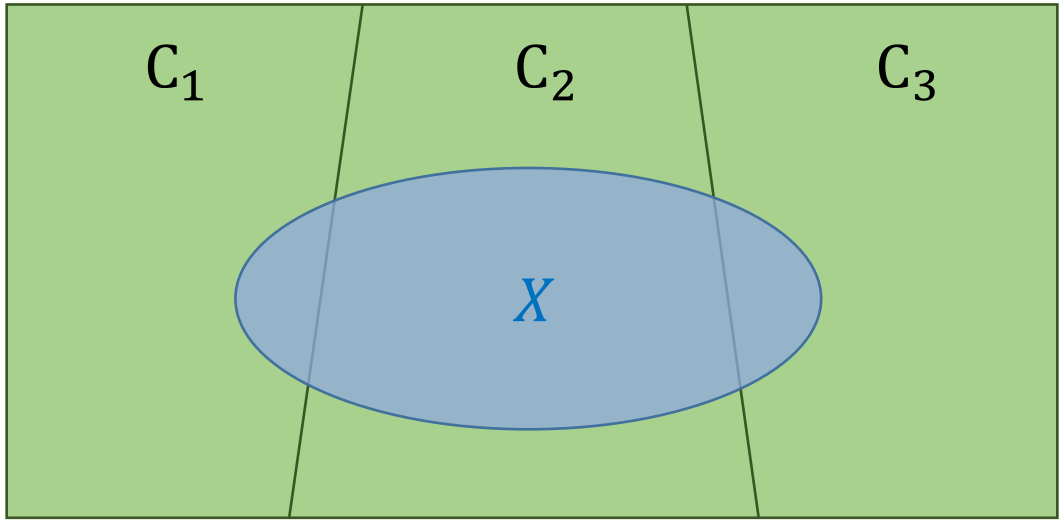
If $C_1, C_2, .. C_n$ are complete system of events, and X is any event with $P(X) \neq 0$

$$P(C_i|X=x) = \frac{P(C_i)P(X=x|C_i)}{P(X=x)} = \frac{P(C_i)P(X=x|C_i)}{\sum_{j=1}^{n} P(C_j)P(X=x|C_j)}, i = 1, 2, \dots, n$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

R_1

R_2



$$P(R_1|S=Yes) = \frac{P(R_1)P(S=Yes|R_1)}{P(S=Yes)}$$

$$= \frac{P(R_1)P(S=Yes|R_1)}{P(R_1)P(S=Yes|R_1) + P(R_2)P(S=Yes|R_2)}$$

$$P(R_2|S=Yes) = \frac{P(R_2)P(S=Yes|R_2)}{P(S=Yes)}$$

$$= \frac{P(R_2)P(S=Yes|R_2)}{P(R_1)P(S=Yes|R_1) + P(R_2)P(S=Yes|R_2)}$$

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SECTION 3
Gaussian Function

SECTION 4
NBC for Continuous RV

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

$$p(res = pass) = \frac{3}{6} = 0.5$$

$$p(res = fail) = \frac{3}{6} = 0.5$$

Bernoulli Naïve Bayes Classifier

❖ Example: One feature

Assume Result is not dependent of Studied

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

R_1

R_2

One feature: Studied

Two classes: Fail and Pass

Let R and S are random variables

$$p(R = r|S = s) = \frac{p(S = s|R = r)p(R = r)}{p(S = s)}$$

$$p(R = R_1\,|S = \text{Yes}) =$$

Bernoulli Naïve Bayes Classifier

❖ Example: One feature

$$p(R=r|S=s) = \frac{p(S=s|R=r)p(R=r)}{p(S=s)}$$

$p(R=R_1|S=yes) = ?$ $\quad$ $p(R=R_2|S=yes) = ?$

S: Studied	R: Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

R_1: Fail

R_2: Pass

$$p(R=R_1|\, S=yes) = \frac{P(R=R_1)P(S=Yes|R=R_1)}{P(S=Yes)} = \frac{P(R=R_1)P(S=Yes|R=R_1)}{P(R=R_1)P(S=Yes|R=R_1)+P(R=R_2)P(S=Yes|R=R_2)}$$

$$p(R=R_2|\, S=yes) = \frac{P(R=R_2)P(S=Yes|R=R_2)}{P(S=Yes)} = \frac{P(R=R_2)P(S=Yes|R=R_2)}{P(R=R_1)P(S=Yes|R=R_1)+P(R=R_2)P(S=Yes|R=R_2)}$$

S Studied	R Result
No	Fail (R_1)
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass (R_2)
No	Fail
Yes	???

$$p(R=R_1|\,S=yes) = \frac{P(R=R_1)P(S=Yes|R=R_1)}{P(S=Yes)}$$

$$p(R=R_2|\,S=yes) = \frac{P(R=R_2)P(S=Yes|R=R_2)}{P(S=Yes)}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

$$p(R=pass) = \frac{3}{6} = \frac{1}{2}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

$$p(R=fail) = \frac{3}{6} = \frac{1}{2}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

$$p(R = pass) = \frac{3}{6} = 0.5$$

$$p(R = fail) = \frac{3}{6} = 0.5$$

another way

$$p(S = yes|R = pass) = \frac{2}{3}$$

$$p(S = yes|R = fail) = \frac{1}{3}$$

$$p(R = R_1|\, S = yes) = \frac{P(R = R_1)P(S = Yes|R = R_1)}{P(S = Yes)}$$

$$p(R = R_2|\, S = yes) = \frac{P(R = R_2)P(S = Yes|R = R_2)}{P(S = Yes)}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

$$p(R = pass) = \frac{3}{6} = \frac{1}{2}$$

$$p(R = fail) = \frac{3}{6} = \frac{1}{2}$$

$$p(S = yes|R = pass) = \frac{2}{3}$$

$$p(S = yes|R = fail) = \frac{1}{3}$$

$$p(S = yes) = p(S = yes|R = pass) * p(R = pass) + p(S = yes|R = fail) * p(R = fail)$$

$$= \frac{2}{3} * \frac{1}{2} + \frac{1}{3} * \frac{1}{2} = \frac{1}{2}$$

$$p(R = R_1 | S = yes) = \frac{P(R = R_1)P(S = Yes|R = R_1)}{P(S = Yes)}$$

$$p(R = R_2 | S = yes) = \frac{P(R = R_2)P(S = Yes|R = R_2)}{P(S = Yes)}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

$$p(R = pass) = \frac{3}{6} = \frac{1}{2}$$

$$p(R = fail) = \frac{3}{6} = \frac{1}{2}$$

$$p(S = yes|R = pass) = \frac{2}{3}$$

$$p(S = yes|R = fail) = \frac{1}{3}$$

$$p(S = yes) = \frac{1}{2}$$

$$p(R = pass \mid S = yes) = \frac{p(S = yes \mid R = pass)p(R = pass)}{p(S = yes)}$$

$$= \frac{2}{3} * \frac{1}{2} * \frac{2}{1} = \frac{2}{3}$$

$$p(R = fail \mid S = yes) = \frac{p(S = yes \mid R = fail)p(R = fail)}{p(S = yes)}$$

$$= \frac{1}{3} * \frac{1}{2} * \frac{2}{1} = \frac{1}{3}$$

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail
Yes	???

$$\mathrm{p}(\mathrm{R} = \mathrm{pass}) = \frac{3}{6} = \frac{1}{2}$$

$$\mathrm{p}(\mathrm{R} = \mathrm{fail}) = \frac{3}{6} = \frac{1}{2}$$

$$\mathrm{p}(\mathrm{S} = \mathrm{yes}|\mathrm{R} = \mathrm{pass}) = \frac{2}{3}$$

$$\mathrm{p}(\mathrm{S} = \mathrm{yes}|\mathrm{R} = \mathrm{fail}) = \frac{1}{3}$$

$$\mathrm{p}(\mathrm{S} = \mathrm{yes}) = \frac{1}{2}$$

$$\mathrm{p}(\mathrm{R} = \mathrm{pass} \mid \mathrm{S} = \mathrm{yes}) = \frac{2}{3} * \frac{1}{2} * \frac{2}{1} = \frac{2}{3}$$

$$\mathrm{p}(\mathrm{R} = \mathrm{fail} \mid \mathrm{S} = \mathrm{yes}) = \frac{1}{3} * \frac{1}{2} * \frac{2}{1} = \frac{1}{3}$$

Assume Result is not dependent of Studied

(5 mins QA and break-time)

Quiz 3: Tính …

$$\mathrm{p}(\mathrm{R} = \mathrm{pass}|S = No)$$

Multiple Features

Bernoulli Naïve Bayes Classifier

❖ Example: Three features

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail
Yes	Yes	Yes	???

R_1

R_2

Three features: Confident, Studied, and Sick

Two classes: Fail and Pass

class=? when Confident=Yes and
Studied=Yes and
Sick=Yes

Let R, Co, St, and Si are random variables

$$p(R|Co, St, Si) = \frac{p(Co, St, Si|R)p(R)}{p(Co, St, Si)}$$

Assume Result is not dependent of other features

NBC assumes that the features are conditionally independent, given the target class

Bernoulli Naïve Bayes Classifier

❖ Example: Three features

$$p(R|Co, St, Si) = \frac{p(Co, St, Si|R)p(R)}{p(Co, St, Si)}$$

$$p(R = R_1|Co = yes, St = yes, Si = yes) = ?$$

$$p(R = R_2|Co = yes, St = yes, Si = yes) = ?$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail
Yes	Yes	Yes	???

$$p(R = pass \mid Co = yes, St = yes, Si = yes) =$$

$$\frac{p(Co = yes, St = yes, Si = yes|res = pass) * p(R = pass)}{p(\ Co = yes, St = yes, Si = yes)}$$

$$p(R = fail \mid Co = yes, St = yes, Si = yes) =$$

$$\frac{p(Co = yes, St = yes, Si = yes|R = fail) * p(R = fail)}{p(\ Co = yes, St = yes, Si = yes)}$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail
Yes	Yes	Yes	???

$$\boxed{p(R = \text{pass}) = \frac{3}{6} = 0.5}$$

$$\boxed{p(R = \text{fail}) = \frac{3}{6} = 0.5}$$

$$p(R = \text{pass} \mid Co = \text{yes}, St = \text{yes}, Si = \text{yes}) =$$

$$\frac{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{pass})\, p(\text{res} = \text{pass})}{p(\, Co = \text{yes}, St = \text{yes}, Si = \text{yes})}$$

$$p(R = \text{fail} \mid Co = \text{yes}, St = \text{yes}, Si = \text{yes}) =$$

$$\frac{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{fail})\, p(R = \text{fail})}{p(\, Co = \text{yes}, St = \text{yes}, Si = \text{yes})}$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail

$$p(R = pass) = \frac{3}{6} = 0.5$$

$$p(R = fail) = \frac{3}{6} = 0.5$$

$$\mathrm{p(Co = yes\ |R = pass)} = \frac{2}{3}$$

$$\mathrm{p(St = yes|R = pass)} = \frac{2}{3}$$

$$\mathrm{p(Si = yes|R = pass)} = \frac{1}{3}$$

$$\mathrm{p(Co = yes, St = yes, Si = yes|R = pass)}$$

$$= \mathrm{p(Co = yes\ |R = pass) * p(St = yes\ |R = pass) * p(Si = yes|R = pass)}$$

$$= \frac{2}{3} * \frac{2}{3} * \frac{1}{3} = \frac{4}{27}$$

features are independent, give Result

$$\mathrm{p(R = pass\ |\ conf = yes, St = yes, Si = yes)} =$$

$$\frac{\mathrm{p(Co = yes, St = yes, Si = yes|R = pass) * p(R = pass)}}{\mathrm{p(\ Co = yes, St = yes, Si = yes)}}$$

$$\mathrm{p(R = fail\ |\ Co = yes, St = yes, Si = yes)} =$$

$$\frac{\mathrm{p(Co = yes, St = yes, Si = yes|R = fail) * p(R = fail)}}{\mathrm{p(Co = yes, St = yes, Si = yes)}}$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail

$$p(R = pass) = \frac{3}{6} = 0.5$$

$$p(R = fail) = \frac{3}{6} = 0.5$$

$$p(Co = yes \mid R = fail) = \frac{1}{3}$$

$$p(St = yes \mid R = fail) = \frac{1}{3}$$

$$p(Si = yes \mid R = fail) = \frac{2}{3}$$

$$p(Co = yes, St = yes, Si = yes \mid res = fail)$$

$$= p(Co = yes \mid res = fail) * p(St = yes \mid res = fail) * p(Si = yes \mid res = fail)$$

$$= \frac{1}{3} * \frac{1}{3} * \frac{2}{3} = \frac{2}{27}$$

features are independent

$$p(R = pass \mid Co = yes, St = yes, Si = yes) = \frac{\boxed{p(Co = yes, St = yes, Si = yes \mid R = pass)} * \boxed{p(R = pass)}}{p(Co = yes, St = yes, Si = yes)}$$

$$p(R = fail \mid Co = yes, St = yes, Si = yes) = \frac{\boxed{p(Co = yes, St = yes, Si = yes \mid R = fail)} * \boxed{p(R = fail)}}{p(Co = yes, St = yes, Si = yes)}$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail

$$p(R = \text{pass}) = \frac{3}{6} = \frac{1}{2} \qquad p(R = \text{fail}) = \frac{3}{6} = \frac{1}{2}$$

$$p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{pass}) = \frac{4}{27}$$

$$p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | res = \text{fail}) = \frac{2}{27}$$

$$p(Co = \text{yes}, St = \text{yes}, Si = \text{yes}) = p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{pass}) * p(R = \text{pass}) + p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{fail}) * p(R = \text{fail})$$

$$= \frac{4}{27} * \frac{1}{2} + \frac{2}{27} * \frac{1}{2} = \frac{6}{27 * 2} = \frac{3}{27} = \frac{1}{9}$$

$$p(R = \text{pass} \mid Co = \text{yes}, St = \text{yes}, Si = \text{yes}) = \frac{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{pass}) * p(R = \text{pass})}{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes})}$$

$$p(R = \text{fail} \mid Co = \text{yes}, St = \text{yes}, Si = \text{yes}) = \frac{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes} | R = \text{fail}) * p(R = \text{fail})}{p(Co = \text{yes}, St = \text{yes}, Si = \text{yes})}$$

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail
Yes	Yes	Yes	???

$$p(R = pass) = \frac{3}{6} = \frac{1}{2} \qquad p(R = fail) = \frac{3}{6} = \frac{1}{2}$$

$$p(Co = yes, St = yes, Si = yes|R = pass) = \frac{4}{27}$$

$$p(Co = yes, St = yes, Si = yes|res = fail) = \frac{2}{27}$$

$$p(Co = yes, St = yes, Si = yes) = \frac{1}{9}$$

$$p(R = pass \mid Co = yes, St = yes, Si = yes)$$

$$= \frac{p(Co = yes, St = yes, Si = yes|R = pass) * p(R = pass)}{p(\ Co = yes, St = yes, Si = yes)} = \frac{4}{27} * \frac{1}{2} * \frac{9}{1} = \frac{2}{3}$$

$$p(R = fail \mid Co = yes, St = yes, Si = yes)$$

$$= \frac{p(Co = yes, St = yes, Si = yes|R = fail) * p(R = fail)}{p(\ Co = yes, St = yes, Si = yes)} = \frac{2}{27} * \frac{1}{2} * \frac{9}{1} = \frac{1}{3}$$

Quiz 4: Tìm mối quan hệ của các biến ngẫu nhiên khi cho cột target là cột Confident. Viết các công thức xác suất mà có thể rút ra được?

Confident	Studied	Sick	Result
Yes	No	No	Fail
Yes	No	Yes	Pass
No	Yes	Yes	Fail
No	Yes	No	Pass
Yes	Yes	No	Pass
No	No	Yes	Fail

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Review

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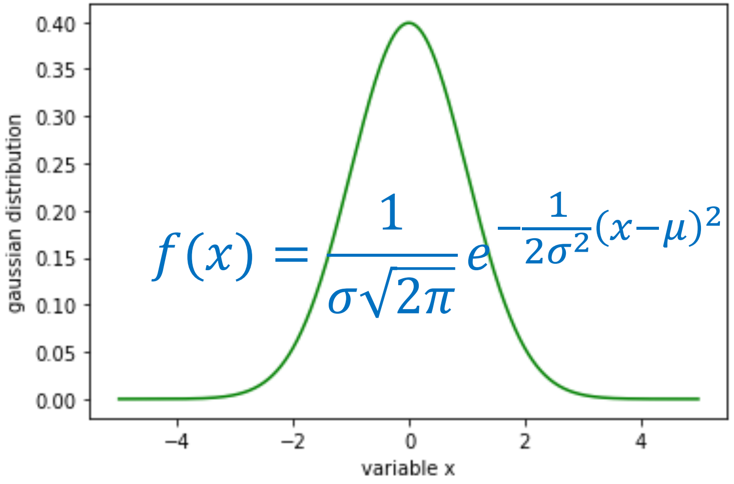
NBC for Discrete RV

SECTION 3

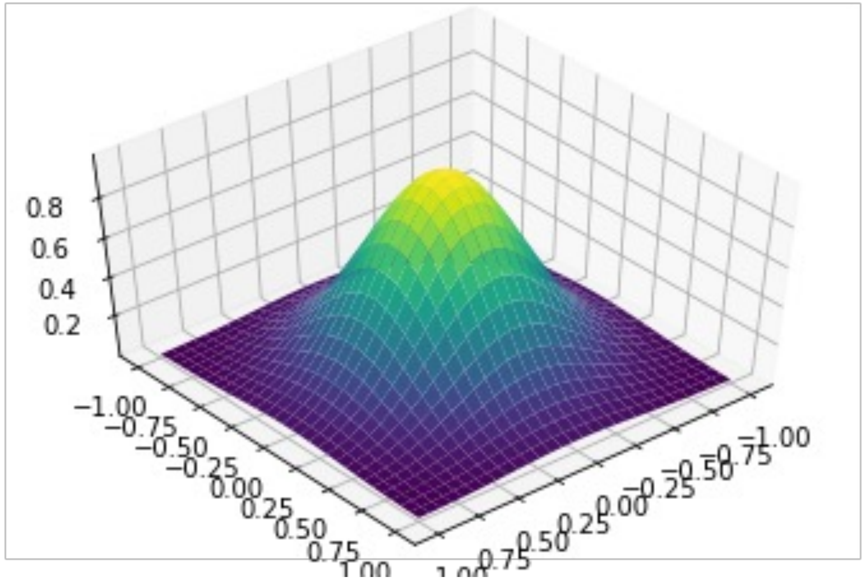
Gaussian Function

SECTION 4

NBC for Continuous RV



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



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Gaussian Function

❖ Definition

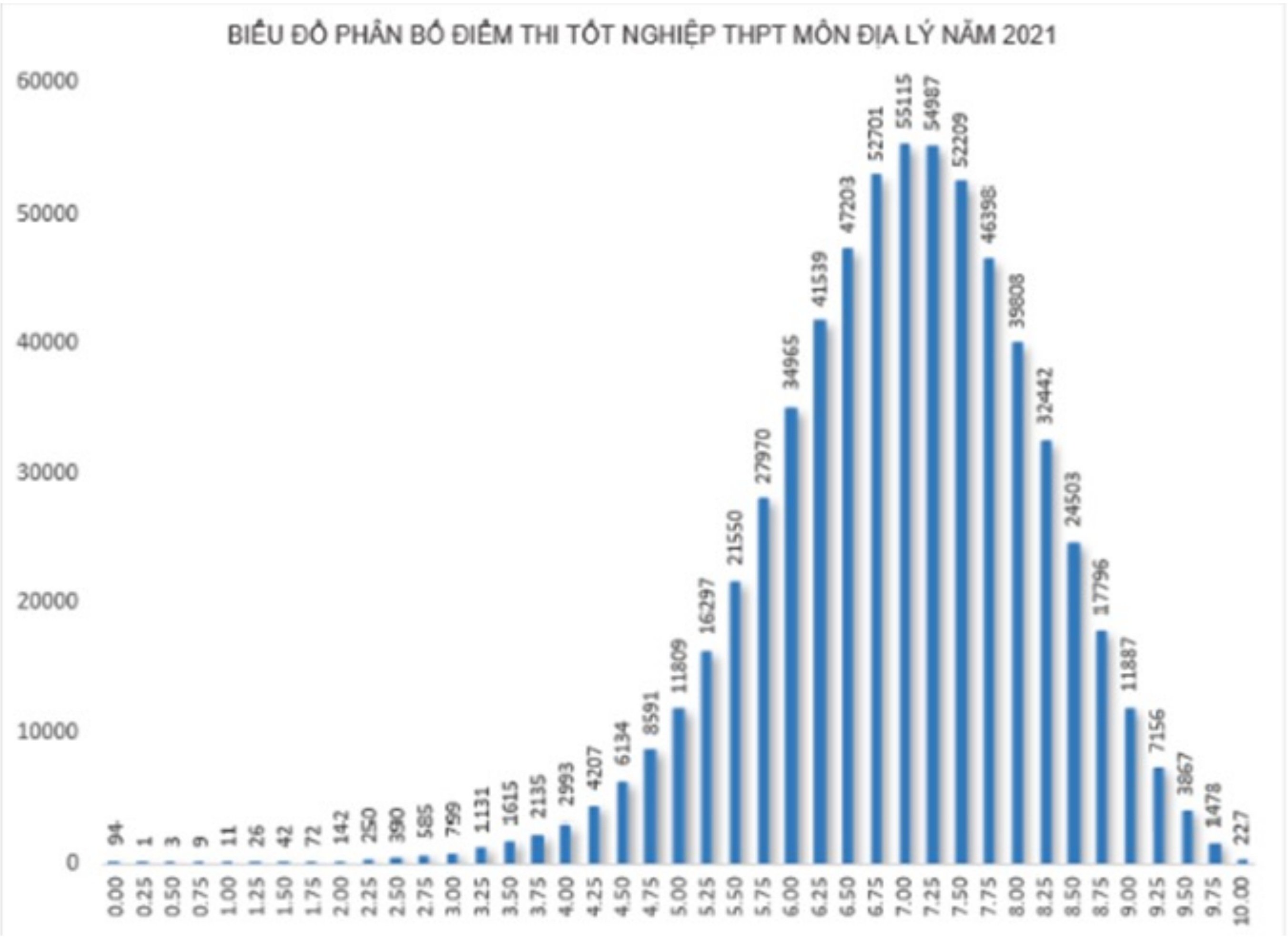
A distribution is simply a collection of data on a variable

Data are arranged in order

Statistics in Plain English

The probability distribution for a random variable describes how the probabilities are distributed over the values of the random variable.

https://www.britannica.com/science/statistics/Random-variables-and-probability-distributions



https://moet.gov.vn/tintuc/Pages/tin-tong-hop.aspx?ItemID=7451

Mean

❖ Definition

Data

$$X = \{X_1, \dots, X_N\}$$

Given the data

$$X = \{2, 8, 5, 4, 1, 4\}$$

$$N = 6$$

Formula

$$\mu_X = \frac{1}{N}\sum_{i=1}^{N} X_i$$

$$\mu_X = \frac{1}{N}\sum_{i-1}^{N} X_i = \frac{1}{6}(2 + 8 + 5 + 4 + 1 + 4)$$

$$= \frac{24}{6} = 4$$

Variance

❖ Definition

Formula:

mean

$$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$

variance

$$var(X) = \frac{1}{n}\sum_{i=1}^{n} (x_i - \mu)^2$$

Example: $X = \{5, 3\ 6, 7, 4\}$

$$\mu = \frac{1}{5}\sum_{i=1}^{n} (5 + 3 + 6 + 7 + 4) = \frac{25}{5} = 5$$

$$var(X) = \frac{1}{5}[(5-5)^2 + (3-5)^2 + (6-5)^2 + (7-5)^2 + (4-5)^2]$$

$$= \frac{1}{5}(0+4+1+4+1)=2$$

Ignore the differences between population and sample, we will discuss them later

Gaussian Function

Công thức

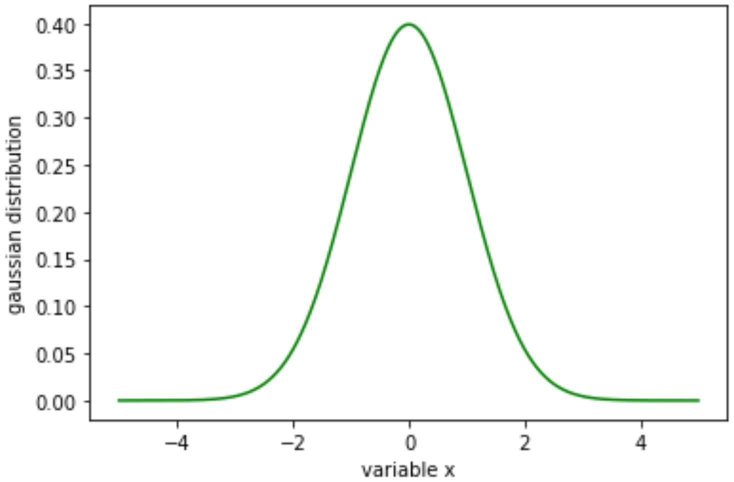
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$-\infty < x < \infty$

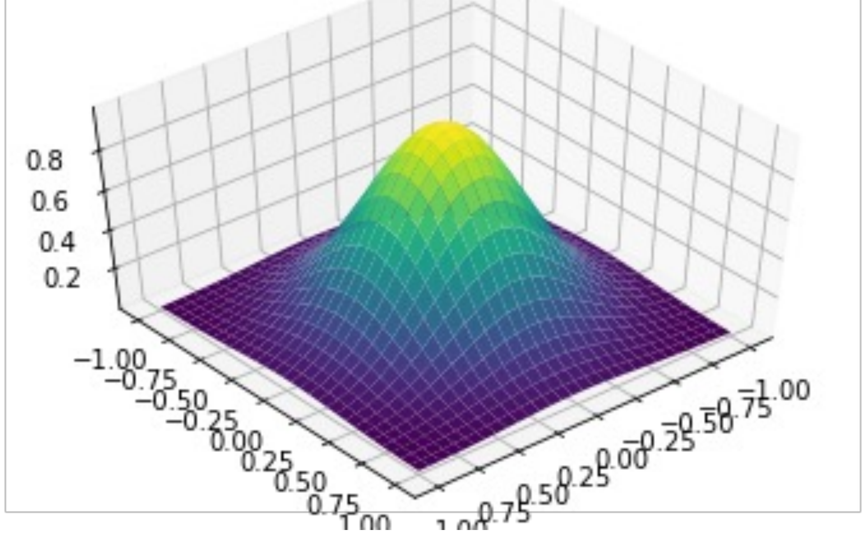
μ: mean

σ^2: variance

Đồ thị Gaussian 1D



Đồ thị Gaussian 2D



Ví dụ

Cho $\mu = 0$ và $\sigma^2 = 1$. Tính giá trị hàm Gaussian với những giá trị x sau

$$f(x=-4) = \frac{1}{1\sqrt{2\pi}} e^{-\frac{1}{2*1}(-4-0)^2} = \frac{e^{-8}}{\sqrt{2\pi}} = \frac{1}{e^8\sqrt{2\pi}} = 1.3\text{e}{-04}$$

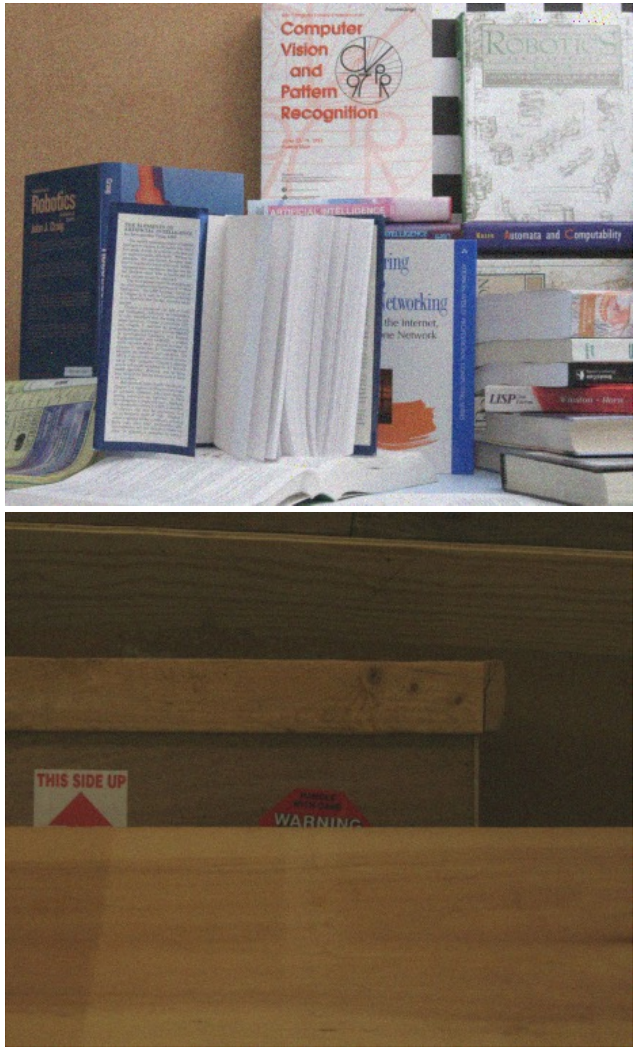
x = [-4 -3 -2 -1 0 1 2 3]

$f(x)$ =[1.3e-04 4.4e-03 5.3e-02 2.4e-01 3.9e-01 2.4e-01 5.3e-02 4.4e-03]

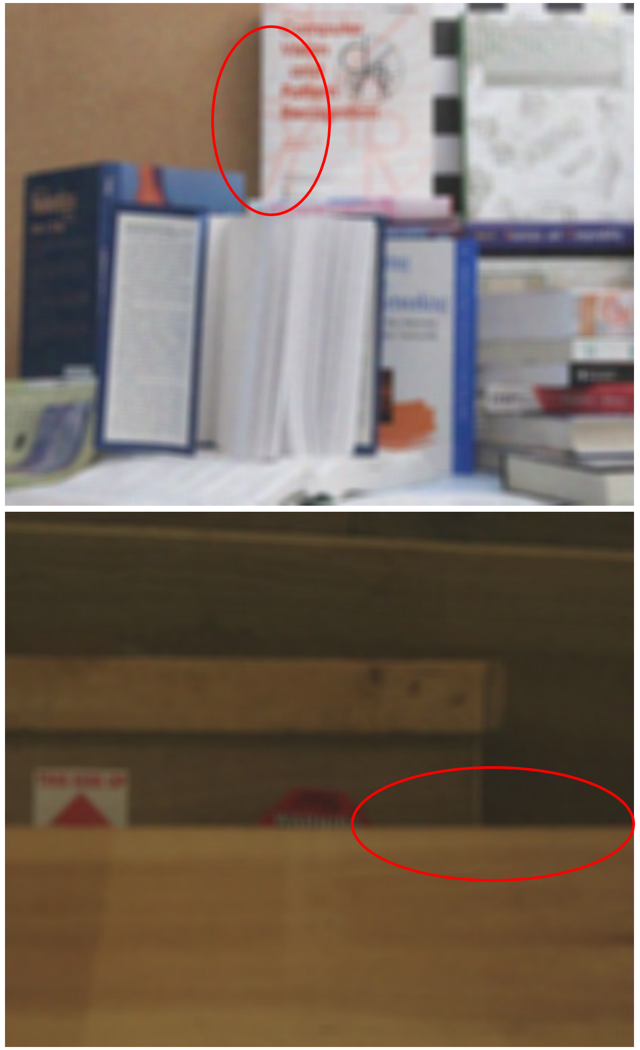
Ứng dụng Gaussian function cho image smoothing

Dùng hàm Gaussian để tính trọng số cho một kernel nxn.

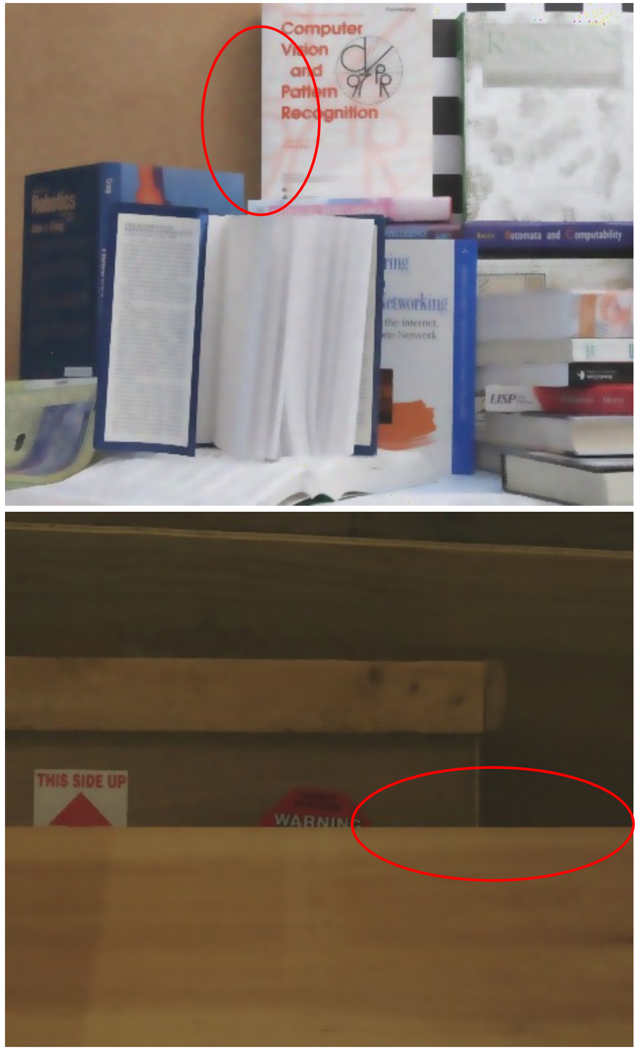
Bilateral filter: dùng thêm thông tin pixel color để hoạt động tốt hơn cho chỗ viền/cạnh.



Ảnh gốc bị nhiễu



Smoothing với trọng số Gausian

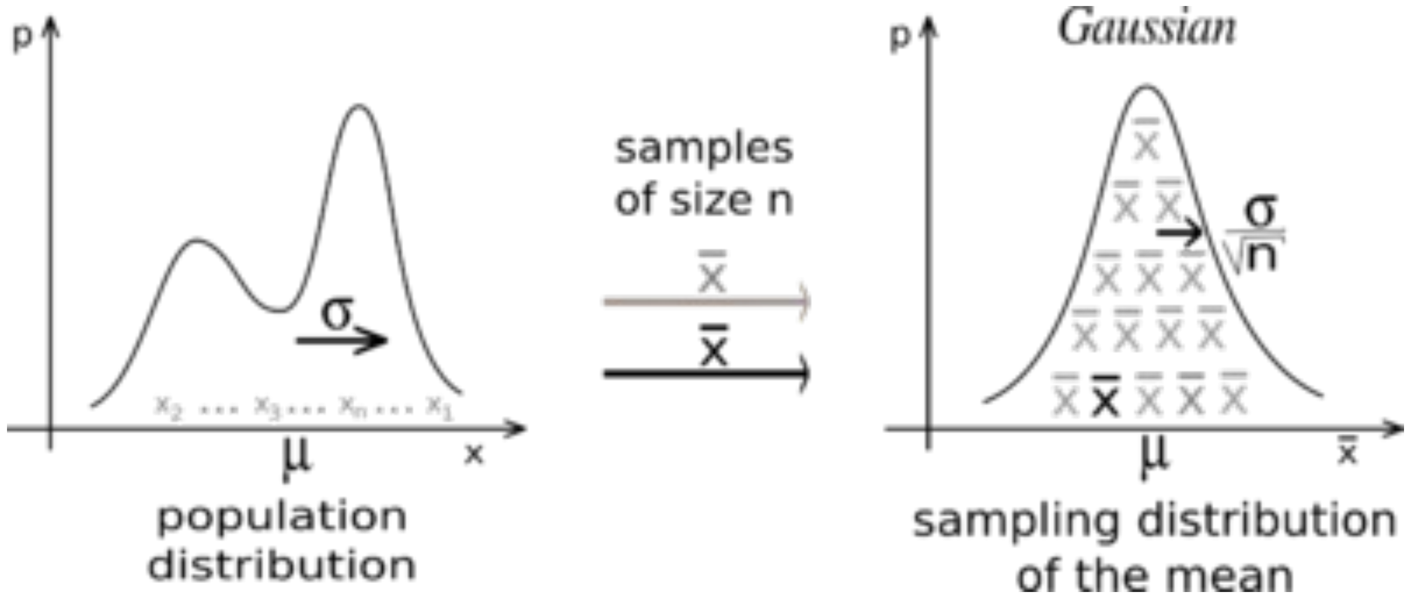


Smoothing với trọng số Gausian+color

Gaussian Function

❖ Central limit theorem

When independent random variables are summed up, their properly normalized sum tends toward a normal distribution even if the original variables themselves are not normally distributed.



https://en.wikipedia.org/wiki/Central_limit_theorem

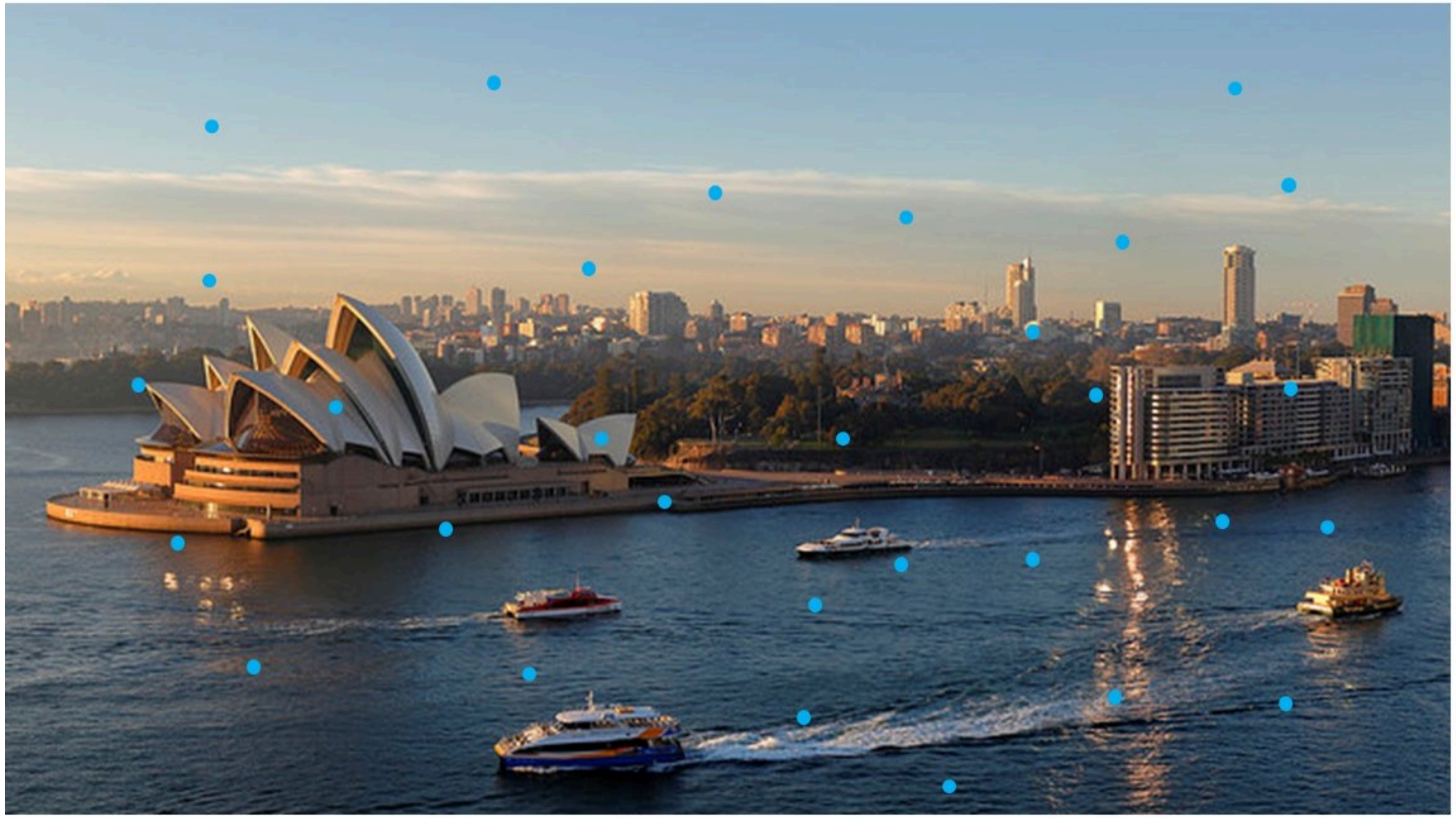
Central Limit Theorem

Randomly selected 30 pixels

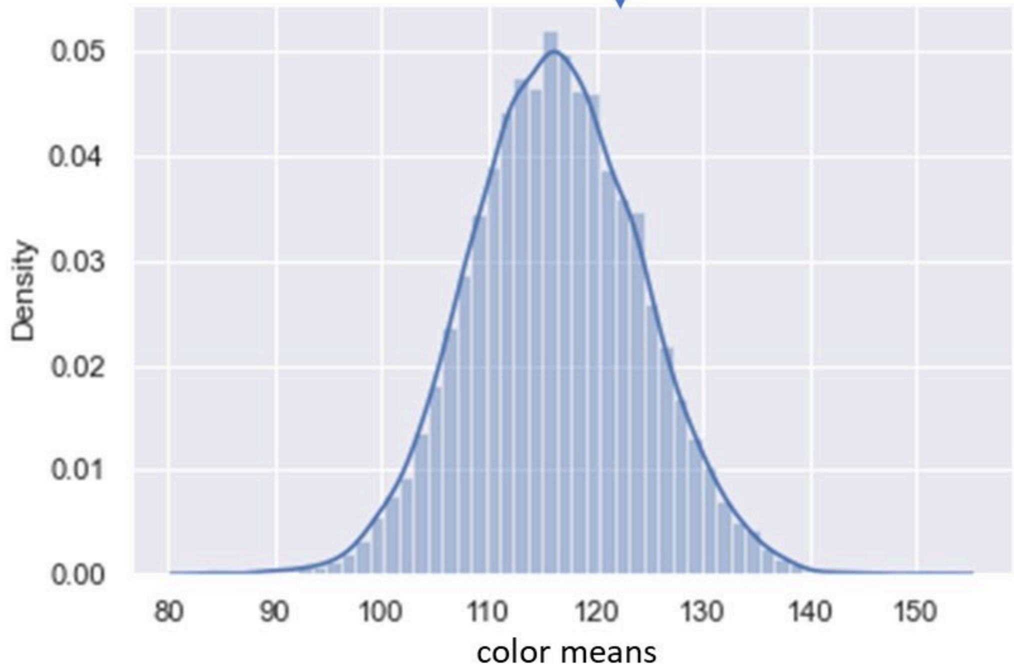
Compute color mean for the 30 pixels

Repeat 10000 times

Histogram of the 10000 color means



Population: pixel colors of the entire image



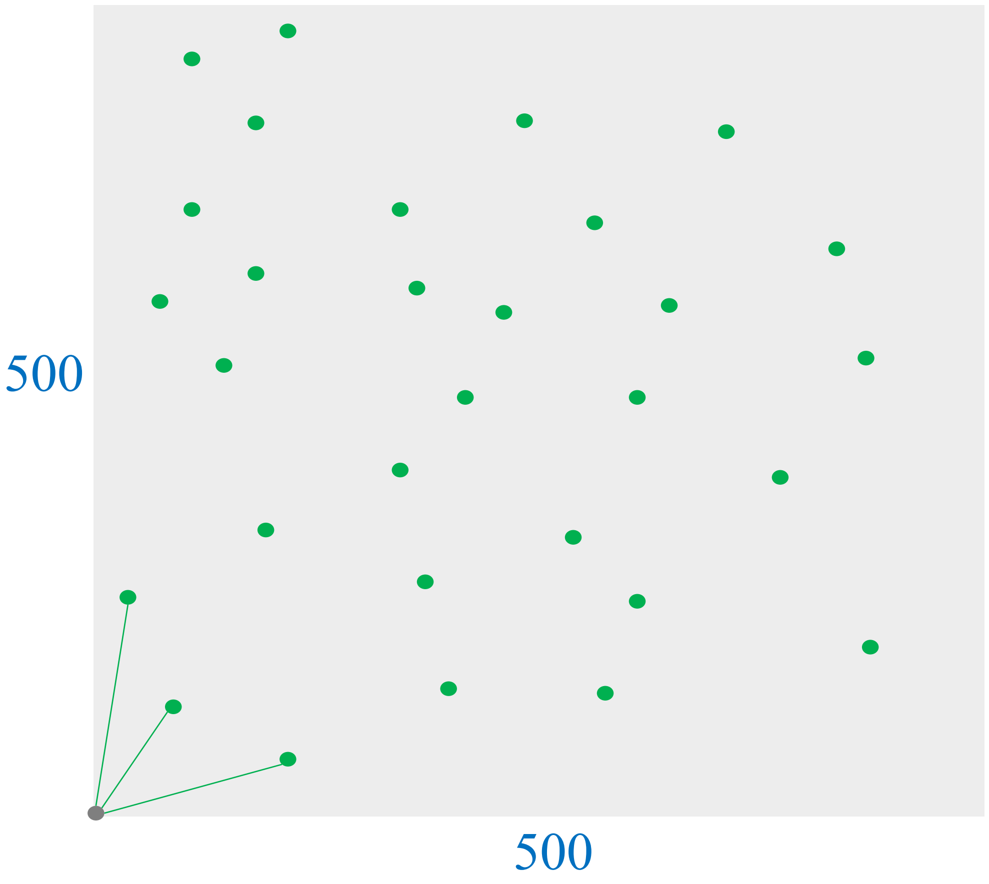
Example

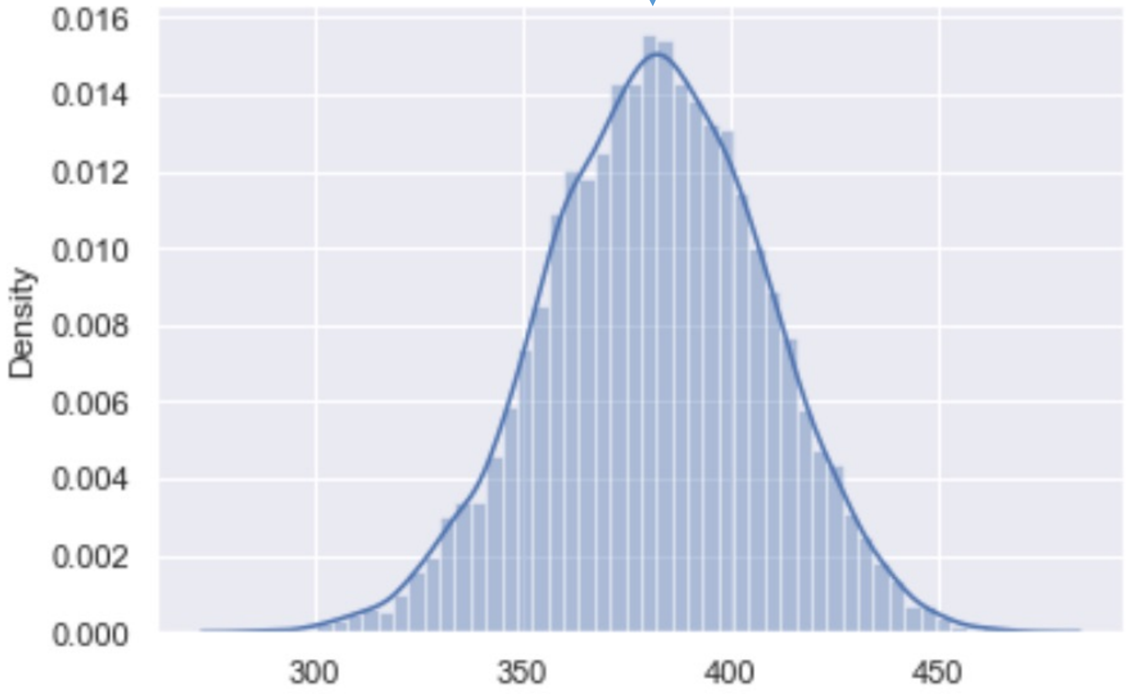
Randomly selected 30 pixels

Compute distance mean

Repeat 10000 times

Histogram of 1000 distance means





Outline

SECTION 1

Review

SECTION 2

NBC for Discrete RV

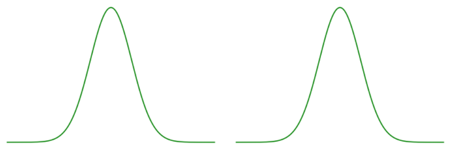
SECTION 3

Gaussian Function

SECTION 4

NBC for Continuous RV

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1



Gaussian Naïve Bayes Classifier

❖ Example: one feature

Assume Category is not dependent of Length

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

One feature: Length (continuous)

Two classes: '0' and '1'

class=? when Length=3.0

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$-\infty < x < \infty$

μ: mean

σ^2: variance

Let C and L are random variables

$$p(C = 0|L = 3.0) = \frac{\boxed{p(L = 3.0|C = 0)} * p(C = 0)}{p(L = 3.0)}$$

$$p(C = 1|L = 3.0) = \frac{\boxed{p(L = 3.0|C = 1)} * p(C = 1)}{p(L = 3.0)}$$

Gaussian Naïve Bayes Classifier

❖ One feature

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

$\mu_0 = 1.52$

$\sigma_0^2 = 0.1416$

$$f(len|Cat=0) = \frac{1}{\sigma_0\sqrt{2\pi}} e^{-\frac{(len-\mu_0)^2}{2\sigma_0^2}}$$

$\mu_1 = 3.88$

$\sigma_1^2 = 0.0776$

$$f(len|Cat=1) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(len-\mu_1)^2}{2\sigma_1^2}}$$

Length	pdf(Length)	Category
1.4	1.007	0
1	0.408	0
1.3	0.893	0
1.9	0.636	0
2	0.469	0
3.8	1.374	1
4.1	1.048	1
3.9	1.428	1
4.2	0.74	1
3.4	0.324	1

$$p(Cat=0|Len=3.0) = \frac{p(Len=3.0|Cat=0) * p(Cat=0)}{p(Len=3.0)}$$

$$p(Cat=1|Len=3.0) = \frac{p(Len=3.0|Cat=1) * p(Cat=1)}{p(Len=3.0)}$$

Gaussian Naïve Bayes Classifier

❖ One feature

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

$\mu_0 = 1.52$

$\sigma_0^2 = 0.1416$

$$f(len|Cat = 0) = \frac{1}{\sigma_0\sqrt{2\pi}} e^{-\frac{(len-\mu_0)^2}{2\sigma_0^2}}$$

$\mu_1 = 3.88$

$\sigma_1^2 = 0.0776$

$$f(len|Cat = 1) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(len-\mu_1)^2}{2\sigma_1^2}}$$

$p(Cat = 0) = 0.5$

$p(Cat = 1) = 0.5$

$$p(Len = 3.0|Cat = 0) = \frac{1}{\sigma_0\sqrt{2\pi}} e^{-\frac{(3.0-\mu_0)^2}{2\sigma_0^2}} = 0.0004638$$

$$p(Len = 3.0|Cat = 1) = \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(3.0-\mu_1)^2}{2\sigma_1^2}} = 0.0097495$$

Gaussian Naïve Bayes Classifier

❖ One feature

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

$$p(\text{Cat} = 0) = 0.5$$

$$p(\text{Cat} = 1) = 0.5$$

$$p(\text{Len} = 3.0|\text{Cat} = 0) = 0.0004638$$

$$p(\text{Len} = 3.0|\text{Cat} = 1) = 0.0097495$$

$$p(\text{Len} = 3.0) = p(\text{Len} = 3.0|\text{Cat} = 0) * p(\text{Cat} = 0) + p(\text{Len} = 3.0|\text{Cat} = 1) * p(\text{Cat} = 1)$$

$$= 0.0004638 * 0.5 + 0.0097495 * 0.5 = 0.005106$$

Gaussian Naïve Bayes Classifier

❖ One feature

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

$$p(Cat = 0) = 0.5$$

$$p(Cat = 1) = 0.5$$

$$p(Len = 3.0|Cat = 0) = 0.0004638$$

$$p(Len = 3.0|Cat = 1) = 0.0097495$$

$$p(Len = 3.0) = 0.005106$$

$$\mathrm{p(Cat = 0|Len = 3.0)} = \frac{\mathrm{p(Len = 3.0|Cat = 0) * p(Cat = 0)}}{\mathrm{p(Len = 3.0)}} = \frac{0.0004638 * 0.5}{0.005106} = 0.0455$$

$$\mathrm{p(Cat = 1|Len = 3.0)} = \frac{\mathrm{p(Len = 3.0|Cat = 1) * p(Cat = 1)}}{\mathrm{p(Len = 3.0)}} = \frac{0.0097495 * 0.5}{0.005106} = 0.9545$$

(3 mins QA and break-time)

Quiz 5: Tại sao không thể dùng cách tính xác suất theo kiểu đếm cho dữ liệu này?

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

Quiz 6: Chọn cột target là cột Length được không?

Length	Category
1.4	0
1	0
1.3	0
1.9	0
2	0
3.8	1
4.1	1
3.9	1
4.2	1
3.4	1

Multiple Features

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

Assume Category is not dependent of Length and/or Width

Assume Length and Width are conditionally independent, given Result

$$p(c|x) = \frac{p(x|c) * p(c)}{p(x)}$$

Two features: Length and Width (continuous)

Two classes: ‘0’ and ‘1’

class=? when Length=4.1 and Width=2.9

$$\mathrm{p(Cat = 0|Len = 4.1, Wid = 2.9)} = \frac{\mathrm{p(Len = 4.1, Wid = 2.9|Cat = 0) * p(Cat = 0)}}{\mathrm{p(Len = 4.1, Wid = 2.9)}}$$

$$\mathrm{p(Cat = 1|Len = 4.1, Wid = 2.9)} = \frac{\mathrm{p(Len = 4.1, Wid = 2.9|Cat = 1) * p(Cat = 1)}}{\mathrm{p(Len = 4.1, Wid = 2.9)}}$$

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

class=? when Length=4.1 and Width=2.9

Let C, W, and L are random variables

$$p(C = 0|L = 4.1, W = 2.9) = \frac{p(L = 4.1, W = 2.9|C = 0) * p(C = 0)}{p(L = 4.1, W = 2.9)}$$

Assume L and W be independent, given C

$$p(L = 4.1, W = 2.9|C = 0) = p(L = 4.1\ |C = 0) * p(W = 2.9|C = 0)$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

$\mu_{0l} = 4.867$ $\sigma_{0l}^2 = 0.078$ $\mu_{0w} = 3.367$ $\sigma_{0w}^2 = 0.095$

$$f(len|Cat = 0) = \frac{1}{\sigma_{0l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{0l})^2}{2\sigma_{0l}^2}} \qquad f(wid|Cat = 0) = \frac{1}{\sigma_{0w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{0w})^2}{2\sigma_{0w}^2}}$$

$\mu_{1l} = 6.216$ $\sigma_{1l}^2 = 0.228$ $\mu_{1w} = 2.916$ $\sigma_{1w}^2 = 0.111$

$$f(len|Cat = 1) = \frac{1}{\sigma_{1l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{1l})^2}{2\sigma_{1l}^2}} \qquad f(wid|Cat = 1) = \frac{1}{\sigma_{1w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{1w})^2}{2\sigma_{1w}^2}}$$

$$p(Cat = 0|Len = 4.1, Wid = 2.9) = \frac{p(Len = 4.1, Wid = 2.9|Cat = 0) * p(Cat = 0)}{p(Len = 4.1, Wid = 2.9)}$$

$$p(Cat = 1|Len = 4.1, Wid = 2.9) = \frac{p(Len = 4.1, Wid = 2.9|Cat = 1) * p(Cat = 1)}{p(Len = 4.1, Wid = 2.9)}$$

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

$$f(\mathrm{Len} = 4.1|\mathrm{Cat} = 0) = \frac{1}{\sigma_{0l}\sqrt{2\pi}} e^{-\frac{(4.1-\mu_{0l})^2}{2\sigma_{0l}^2}} = 0.0342$$

$$f(\mathrm{Wid} = 2.9|\mathrm{Cat} = 0) = \frac{1}{\sigma_{0w}\sqrt{2\pi}} e^{-\frac{(2.9-\mu_{0w})^2}{2\sigma_{0w}^2}} = 0.4129$$

$$f(\mathrm{Len} = 4.1|\mathrm{Cat} = 1) = \frac{1}{\sigma_{1l}\sqrt{2\pi}} e^{-\frac{(4.1-\mu_{1l})^2}{2\sigma_{1l}^2}} = 4e-5$$

$$f(\mathrm{Wid} = 2.9|\mathrm{Cat} = 1) = \frac{1}{\sigma_{1w}\sqrt{2\pi}} e^{-\frac{(2.9-\mu_{1w})^2}{2\sigma_{1w}^2}} = 1.1938$$

$$p(\mathrm{Len} = 4.1, \mathrm{Wid} = 2.9|\mathrm{Cat} = 0) = p(\mathrm{Len} = 4.1\ |\mathrm{Cat} = 0) * p(\mathrm{Wid} = 2.9|\mathrm{Cat} = 0)$$
$$= 0.0342 * 0.4129 = 0.01412$$

$$p(\mathrm{Len} = 4.1, \mathrm{Wid} = 2.9|\mathrm{Cat} = 1) = p(\mathrm{Len} = 4.1\ |\mathrm{Cat} = 1) * p(\mathrm{Wid} = 2.9|\mathrm{Cat} = 1)$$
$$= 0.00004 * 1.1938 = 0.000047$$

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 0) = 0.01412$$

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 1) = 0.000047$$

$$p(\text{Cat} = 0) = 0.5$$

$$p(\text{Cat} = 1) = 0.5$$

$$\begin{aligned} p(\text{Len} = 4.1, \text{Wid} = 2.9) &= p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 0) p(\text{Cat} = 0) \\ &+ p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 1) p(\text{Cat} = 1) \\ &= 0.01412 * 0.5 + 0.000047 * 0.5 = 0.00708 \end{aligned}$$

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 0) = 0.01412$$

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 1) = 0.000047$$

$$p(\text{Cat} = 0) = 0.5$$

$$p(\text{Cat} = 1) = 0.5$$

$$p(\text{Len} = 4.1, \text{Wid} = 2.9) = 0.00708$$

$$p(\text{Cat} = 0 | \text{Len} = 4.1, \text{Wid} = 2.9) = \frac{p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 0) * p(\text{Cat} = 0)}{p(\text{Len} = 4.1, \text{Wid} = 2.9)}$$

$$= \frac{0.01412 * 0.5}{0.00708} = 0.9962$$

Gaussian Naïve Bayes Classifier

❖ Example: two features

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 0) = 0.01412$$

$$p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 1) = 0.000047$$

$$p(\text{Cat} = 0) = 0.5$$

$$p(\text{Cat} = 1) = 0.5$$

$$p(\text{Len} = 4.1, \text{Wid} = 2.9) = 0.00708$$

$$p(\text{Cat} = 0 | \text{Len} = 4.1, \text{Wid} = 2.9) = 0.9962$$

$$p(\text{Cat} = 1 | \text{Len} = 4.1, \text{Wid} = 2.9) = \frac{p(\text{Len} = 4.1, \text{Wid} = 2.9 | \text{Cat} = 1) * p(\text{Cat} = 1)}{p(\text{Len} = 4.1, \text{Wid} = 2.9)}$$

$$= \frac{0.000047 * 0.5}{0.00708} = 0.0038$$

Quiz 7: Giải bài này thế nào?

Europe	Length	Category
Yes	1,4	0
No	1	0
No	1,3	0
Yes	1,9	0
Yes	2	0
No	3,8	1
Yes	4,1	1
No	3,9	1
Yes	4,2	1
No	3,4	1

Multiple Features
Multiple Classes
(Homework)

Gaussian Naïve Bayes Classifier

❖ Three classes

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$-\infty < x < \infty$$

μ: mean

σ^2: variance

$$p(A|B) = \frac{p(B|A) * p(A)}{p(B)}$$

Two features: Length and Width (continuous)

Three classes: ‘0’, ‘1’, and ‘2’

class=? when Length=5.2 and Width=2.4

$$\mathrm{p(Cat = c|Len = 5.2, Wid = 2.4)}$$

$$= \frac{\boxed{\mathrm{p(Len = 5.2, Wid = 2.4|Cat = c)}} * \mathrm{p(Cat = c)}}{\mathrm{p(Len = 5.2, Wid = 2.4)}}$$

$$\mathrm{p(Cat = 0|Len = 5.2, Wid = 2.4)}$$

$$\mathrm{p(Cat = 1|Len = 5.2, Wid = 2.4)}$$

$$\mathrm{p(Cat = 2|Len = 5.2, Wid = 2.4)}$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$\mu_{0l} = 4.867$ $\sigma_{0l}^2 = 0.078$ $\mu_{0w} = 3.367$ $\sigma_{0w}^2 = 0.095$

$$f(len|Cat = 0) = \frac{1}{\sigma_{0l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{0l})^2}{2\sigma_{0l}^2}} \qquad f(wid|Cat = 0) = \frac{1}{\sigma_{0w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{0w})^2}{2\sigma_{0w}^2}}$$

$\mu_{1l} = 6.216$ $\sigma_{1l}^2 = 0.228$ $\mu_{1w} = 2.916$ $\sigma_{1w}^2 = 0.111$

$$f(len|Cat = 1) = \frac{1}{\sigma_{1l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{1l})^2}{2\sigma_{1l}^2}} \qquad f(wid|Cat = 1) = \frac{1}{\sigma_{1w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{1w})^2}{2\sigma_{1w}^2}}$$

$\mu_{2l} = 6.216$ $\sigma_{2l}^2 = 0.228$ $\mu_{2w} = 2.916$ $\sigma_{2w}^2 = 0.111$

$$f(len|Cat = 2) = \frac{1}{\sigma_{2l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{2l})^2}{2\sigma_{2l}^2}} \qquad f(wid|Cat = 2) = \frac{1}{\sigma_{2w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{2w})^2}{2\sigma_{2w}^2}}$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$$f(Len = 5.2|Cat = 0) = \frac{1}{\sigma_{0l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{0l})^2}{2\sigma_{0l}^2}} = 0.7023$$

$$f(Wid = 2.4|Cat = 0) = \frac{1}{\sigma_{0w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{0w})^2}{2\sigma_{0w}^2}} = 0.0097$$

$$f(Len = 5.2|Cat = 1) = \frac{1}{\sigma_{1l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{1l})^2}{2\sigma_{1l}^2}} = 0.0866$$

$$f(Wid = 2.4|Cat = 1) = \frac{1}{\sigma_{1w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{1w})^2}{2\sigma_{1w}^2}} = 0.3606$$

$$f(Len = 5.2|Cat = 2) = \frac{1}{\sigma_{2l}\sqrt{2\pi}} e^{-\frac{(len-\mu_{2l})^2}{2\sigma_{2l}^2}} = 0.6785$$

$$f(Wid = 2.4|Cat = 2) = \frac{1}{\sigma_{2w}\sqrt{2\pi}} e^{-\frac{(wid-\mu_{2w})^2}{2\sigma_{2w}^2}} = 0.1839$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$f(\text{Len} = 5.2|\text{Cat} = 0) = 0.7023$ $\quad$ $f(\text{Wid} = 2.4|\text{Cat} = 0) = 0.0097$

$f(\text{Len} = 5.2|\text{Cat} = 1) = 0.0866$ $\quad$ $f(\text{Wid} = 2.4|\text{Cat} = 1) = 0.3606$

$f(\text{Len} = 5.2|\text{Cat} = 2) = 0.6785$ $\quad$ $f(\text{Wid} = 2.4|\text{Cat} = 2) = 0.1839$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0)$$

$$= p(\text{Len} = 5.2\ |\text{Cat} = 0) * p(\text{Wid} = 2.4|\text{Cat} = 0)$$

$$= 0.7023 * 0.0097 = 0.00681$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$$f(\text{Len} = 5.2|\text{Cat} = 0) = 0.7023 \qquad f(\text{Wid} = 2.4|\text{Cat} = 0) = 0.0097$$

$$f(\text{Len} = 5.2|\text{Cat} = 1) = 0.0866 \qquad f(\text{Wid} = 2.4|\text{Cat} = 1) = 0.3606$$

$$f(\text{Len} = 5.2|\text{Cat} = 2) = 0.6785 \qquad f(\text{Wid} = 2.4|\text{Cat} = 2) = 0.1839$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0) = 0.00681$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 1)$$

$$= p(\text{Len} = 5.2\ |\text{Cat} = 1) * p(\text{Wid} = 2.4|\text{Cat} = 1)$$

$$= 0.0866 * 0.3606 = 0.03122$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$f(Len = 5.2|\mathrm{Cat} = 0) = 0.7023$

$f(Wid = 2.4|\mathrm{Cat} = 0) = 0.0097$

$f(Len = 5.2|\mathrm{Cat} = 1) = 0.0866$

$f(Wid = 2.4|\mathrm{Cat} = 1) = 0.3606$

$f(Len = 5.2|\mathrm{Cat} = 2) = 0.6785$

$f(Wid = 2.4|\mathrm{Cat} = 2) = 0.1839$

$p(Len = 5.2, Wid = 2.4|Cat = 0) = 0.00681$

$p(Len = 5.2, Wid = 2.4|Cat = 1) = 0.03122$

$$\begin{aligned}\mathrm{p(Len = 5.2, Wid = 2.4|Cat = 2)} \\ &= \mathrm{p(Len = 5.2\ |Cat = 2) * p(Wid = 2.4|Cat = 2)} \\ &= 0.6785 * 0.1839 = 0.1247\end{aligned}$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$f(Len = 5.2|Cat = 0) = 0.7023$ $\quad f(Wid = 2.4|Cat = 0) = 0.0097$

$f(Len = 5.2|Cat = 1) = 0.0866$ $\quad f(Wid = 2.4|Cat = 1) = 0.3606$

$f(Len = 5.2|Cat = 2) = 0.6785$ $\quad f(Wid = 2.4|Cat = 2) = 0.1839$

$$p(Len = 5.2, Wid = 2.4|Cat = 0) = 0.00681$$

$$p(Len = 5.2, Wid = 2.4|Cat = 1) = 0.03122$$

$$p(Len = 5.2, Wid = 2.4|Cat = 2) = 0.1247$$

$p(Cat = 0) = 0.333$

$p(Cat = 1) = 0.333$

$p(Cat = 2) = 0.334$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$f(\text{Len} = 5.2|\text{Cat} = 0) = 0.7023$ $\quad f(\text{Wid} = 2.4|\text{Cat} = 0) = 0.0097$

$f(\text{Len} = 5.2|\text{Cat} = 1) = 0.0866$ $\quad f(\text{Wid} = 2.4|\text{Cat} = 1) = 0.3606$

$f(\text{Len} = 5.2|\text{Cat} = 2) = 0.6785$ $\quad f(\text{Wid} = 2.4|\text{Cat} = 2) = 0.1839$

$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0) = 0.00681$ $\quad p(\text{Cat} = 0) = 0.333$

$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 1) = 0.03122$ $\quad p(\text{Cat} = 1) = 0.333$

$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 2) = 0.1247$ $\quad p(\text{Cat} = 2) = 0.334$

$$\begin{aligned}
p(\text{Len} = 5.2, \text{Wid} = 2.4) &= p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0) * p(\text{Cat} = 0) \\
&+ p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 1) * p(\text{Cat} = 1) \\
&+ p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 2) * p(\text{Cat} = 2) \\
&= 0.00681 * 0.333 + 0.03122 * 0.333 + 0.1247 * 0.334 = 0.0543
\end{aligned}$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$$f(\text{Len} = 5.2|\text{Cat} = 0) = 0.7023 \qquad f(\text{Wid} = 2.4|\text{Cat} = 0) = 0.0097$$

$$f(\text{Len} = 5.2|\text{Cat} = 1) = 0.0866 \qquad f(\text{Wid} = 2.4|\text{Cat} = 1) = 0.3606$$

$$f(\text{Len} = 5.2|\text{Cat} = 2) = 0.6785 \qquad f(\text{Wid} = 2.4|\text{Cat} = 2) = 0.1839$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0) = 0.00681 \qquad p(\text{Cat} = 0) = 0.333$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 1) = 0.03122 \qquad p(\text{Cat} = 1) = 0.333$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 2) = 0.1247 \qquad p(\text{Cat} = 2) = 0.334$$

$$p(\text{Len} = 5.2, \text{Wid} = 2.4) = 0.0543$$

$$p(\text{Cat} = 0|\text{Len} = 5.2, \text{Wid} = 2.4)$$

$$= \frac{p(\text{Len} = 5.2, \text{Wid} = 2.4|\text{Cat} = 0) * p(\text{Cat} = 0)}{p(\text{Len} = 5.2, \text{Wid} = 2.4)} = \frac{0.00681 * 0.333}{0.0543} = 0.041$$

Gaussian Naïve Bayes Classifier

❖ Example

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1
5.6	2.2	2
5.1	1.5	2
5.6	1.4	2
5.9	2.1	2
5.3	1.8	2
5.7	1.9	2

$f(\mathrm{Len} = 5.2|\mathrm{Cat} = 0) = 0.7023$ $\quad f(\mathrm{Wid} = 2.4|\mathrm{Cat} = 0) = 0.0097$

$f(\mathrm{Len} = 5.2|\mathrm{Cat} = 1) = 0.0866$ $\quad f(\mathrm{Wid} = 2.4|\mathrm{Cat} = 1) = 0.3606$

$f(\mathrm{Len} = 5.2|\mathrm{Cat} = 2) = 0.6785$ $\quad f(\mathrm{Wid} = 2.4|\mathrm{Cat} = 2) = 0.1839$

$p(\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4|\mathrm{Cat} = 0) = 0.00681$ $\quad p(\mathrm{Cat} = 0) = 0.333$

$p(\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4|\mathrm{Cat} = 1) = 0.03122$ $\quad p(\mathrm{Cat} = 1) = 0.333$

$p(\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4|\mathrm{Cat} = 2) = 0.1247$ $\quad p(\mathrm{Cat} = 2) = 0.334$

$p(\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4) = 0.0543$

$p(\mathrm{Cat} = 0|\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4) = 0.042$

$p(\mathrm{Cat} = 1|\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4) = 0.192$

$\boxed{p(\mathrm{Cat} = 2|\mathrm{Len} = 5.2, \mathrm{Wid} = 2.4) = 0.766}$

Summary

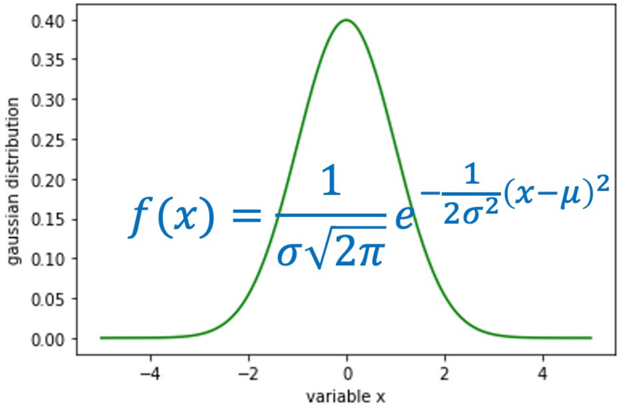
NBC for Discrete RV

Studied	Result
No	Fail
No	Pass
Yes	Fail
Yes	Pass
Yes	Pass
No	Fail

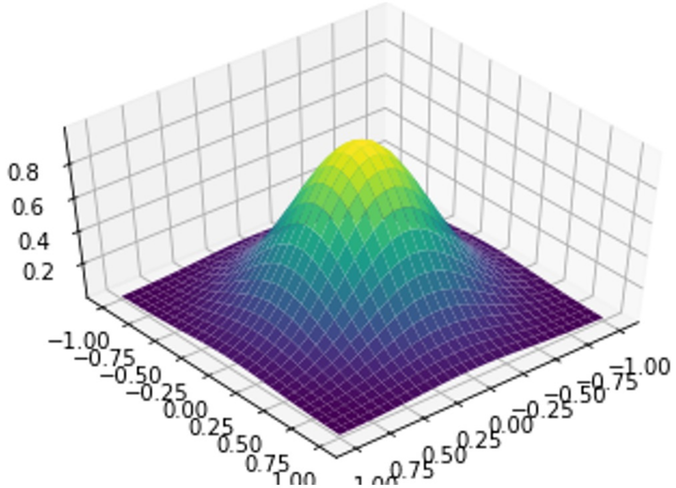
$$p(res = pass) = \frac{3}{6} = 0.5$$

$$p(res = fail) = \frac{3}{6} = 0.5$$

Gaussian Function



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$



NBC for Cont. RV

Length	Width	Category
4.9	3	0
4.7	3.2	0
4.6	3.1	0
5	3.6	0
5.4	3.9	0
4.6	3.4	0
6.4	3.2	1
6.9	3.1	1
5.5	2.3	1
6.5	2.8	1
5.7	2.8	1
6.3	3.3	1